Predicting Forex Prices: An Evaluation of Long Short-Term Memory, XGBoost and Transformer Architectures

1st Yash Dave
School of Computing
Unitec Institute of Technology
Auckland, New Zealand
yashacnz@gmail.com

2nd Soheil Varastehpour
School of Computing
Unitec Institute of Technology
Auckland, New Zealand
spour@unitec.ac.nz

3rd Masoud Shakiba
School of Computing
Unitec Institute of Technology
Auckland, New Zealand
mshakiba@unitec.ac.nz

2025 5th International Conference on Advances in Electrical, Electronics and Computing Technology (EECT) | 979-8-3315-4154-5/25/$31.00 ©2025 IEEE | DOI: 10.1109/EECT64505.2025.10966964

***Abstract*—This research explores the effectiveness of various machine learning and deep learning models in predicting Forex prices using technical and fundamental indicators. Given the Forex market's unpredictability, accurate models are crucial for informed trading decisions and academic research. We use technical, fundamental, and combined indicators to compare models like Long Short-Term Memory, XGBoost, and Transformer-based architectures across different settings. An ensemble model is also introduced to balance accuracy and reliability. The analysis focuses on currency pairs like EUR/USD, NZD/USD, EUR/NZD, and AUD/NZD over a daily timeframe. We use metrics such as Mean Absolute Error, Mean Absolute Percentage Error, Root Mean Square Error, and R-squared to evaluate performance. Findings suggest that only some models excel in all scenarios, underscoring the need for model selection based on market conditions. This paper provides valuable insights into the strengths and weaknesses of different models for Forex price prediction, benefiting both academics and practitioners in finance.**

***Index Terms*—Forex Prediction, Deep Learning, LSTM, XGBoost, Transformer**

I. Introduction

The foreign exchange (Forex) market, known for its liquidity and continuous operation, is critical for global currency trading. Accurate forex price prediction is essential for investors and analysts, as it directly influences trading decisions. This challenge has driven the exploration of machine learning (ML) and deep learning (DL) models to forecast market trends more effectively [1]. Recent studies highlight deep learning's effectiveness in capturing complex, non-linear patterns in Forex price movements [2].

This research leverages technical models (TM) and fundamental models (FM) for market prediction. TM analyses historical market data through indicators like OPEN, HIGH, LOW, and CLOSE prices, Average True Range (ATR), Rate of Change (ROC_7), Relative Strength Index (RSI_14), Exponential Moving Average (EMA_50), and Bollinger Bands. FM assesses the intrinsic value of currencies using economic indicators like commodity prices, inflation, interest rates, GDP, and major stock indices. This paper uses technical and fundamental indicators to compare ML and DL models for forex prediction. It introduces an ensemble model that combines the strengths of individual models through a weighted average method to improve accuracy. By evaluating these models across major currency pairs, this study offers insights into their strengths and weaknesses, aiding in informed trading decisions in a volatile market environment.

II. Literature Review

Forex prediction has garnered significant attention in financial research, with various studies employing ML and DL models to enhance prediction accuracy. Notably, research employing recurrent neural network (RNN) architectures, including Long Short-Term Memory (LSTM) units, Gated Recurrent Units (GRUs), and bidirectional RNNs, has shown promise in forecasting forex exchange rates. For instance, a study on the EUR/USD exchange rate from 2000 to 2019 utilized open, high, low, and close prices along with technical indicators like the Moving Average Convergence Divergence (MACD), RSI, and stochastic oscillator [3], [4]. However, the challenge of overfitting, where models excessively adapt to training data at the expense of generalization, remains a critical concern not thoroughly addressed in some studies. Further research integrating LSTM models with both technical and macroeconomic indicators attempted to forecast the directional movement of the EUR/USD exchange rate [5]. This approach highlights the importance of selecting relevant features, as including technical indicators and macroeconomic data like GDP, inflation, and unemployment rates can lead to redundancy or negative impacts on model performance if not appropriately managed. Deep learning models, such as LSTM, Convolutional Neural Networks (CNNs), and Multilayer Perceptrons (MLPs), have also been applied to forex prediction tasks. A notable study used EUR/USD exchange rate data from 2000 to 2016, focusing solely on price data without considering the model's long-term performance or the impact of market volatility on predictions [6].

Two recent surveys provide comprehensive overviews of deep learning and machine learning approaches applied to Forex prediction. The first review various deep learning models applied to Forex and stock prediction, evaluating datasets, variables, and performance results across multiple studies [7]. This survey underlines the growing preference for deep learning models, particularly RNNs and Transformers, for time-series forecasting tasks in finance. The second provides a meta-analysis of machine learning approaches, analyzing models, datasets, and features used in Forex forecasting [8]. This analysis identifies gaps in model performance due to dataset limitations and varying feature choices, which influence the predictive effectiveness of ML models.

Existing studies on Forex price prediction primarily focus on using either technical indicators or fundamental economic factors individually [3], [4]. While models such as ARIMA and basic RNNs have shown some predictive success, they often struggle with the complex, non-linear dynamics of the Forex market, especially when handling non-stationary data [6]. Moreover, many studies must account for multi-currency comparisons or the full breadth of economic cycles, leading to limited generalizability. This research addresses these gaps by combining technical and fundamental indicators within a novel ensemble framework (LSTM, XGBoost, Transformer), offering a more robust predictive tool tailored to the unique characteristics of the Forex market. While previous research has combined these indicators, few studies incorporate a broader range of financial data, such as transport activity, commodity prices, and industry-specific metrics that capture unique economic dynamics and improve predictive insight.

A. Role of Technical and Fundamental Indicators

TABLE I
ANALYSIS OF FOREX PAIRS WITH TECHNICAL AND FUNDAMENTAL INDICATORS

Forex Pair	Technical Indicators	Fundamental Indicators
EUR/USD	OPEN, HIGH, LOW, CLOSE, ATR, ROC_7, RSI_14, EMA_50, MA_20, Upper/Lower Band	Gold, DXY, Oil, S&P, EU50, EU/US M2, EUR/USD Inflation Rate, Interest Rate, USBOAT, EURBOT, EUUR, USUR, GDP (US/EU)
NZD/USD	Similar technical setup to EUR/USD	NZX50, NZD Inflation Rate, Interest Rate, NZGDTPI, NZTA
EUR/NZD	Technical setup similar to EUR/USD	Adjustments for NZD-specific factors; EU M3, NZD M3
AUD/NZD	Similar technical setup to EUR/USD	AUD/NZD M3, Inflation Rate, Interest Rate, NZTA, NZBOT, AUBOT, NZUR, AUUR, GDP (NZ/AU)

In the search to enhance forex price prediction accuracy, the integration of TM and FM indicators plays a pivotal role, as evidenced by recent studies employing a spectrum ML and DL models. These investigations have explored various modeling techniques, including LSTM, XGBoost, and Transformer-based architectures, applying them across different configurations of TM and FM indicators [9], [10]. This section underscores the importance of these analytical approaches in the context of forex markets, leveraging the detailed analysis of models utilizing a comprehensive set of indicators for each key currency pair. As presented in Table I, the combined set of technical and fundamental indicators used for analysing the EUR/USD, NZD/USD, EUR/NZD, and AUD/NZD currency pairs form the data-driven foundation of our nuanced approach to understanding and predicting market behaviours.

a) Technical Indicators: The selected technical indicators are crucial in analysing short-term price movements and market dynamics. The ATR measures market volatility by quantifying the degree of price variation, providing insights into the stability of a currency pair. The ROC indicator detects momentum by comparing the current price with prices a specific number of periods ago, offering a clear picture of trend direction and speed. The RSI is critical for identifying overbought or oversold conditions, serving as a precursor to potential market reversals. Additionally, various moving averages, such as the EMA are used to smooth price data to identify ongoing market trends and potential points of resistance and support.

b) Fundamental Indicators: Fundamental indicators contain economic factors affecting currency strength and market sentiment. Key indicators include monetary aggregates like M2 and M3. M2, which includes liquid assets, forecasts future inflation and economic trends, while M3, encompassing large deposits and institutional funds, signals overall economic health and potential monetary policy shifts.Interest rates are pivotal as they influence forex rates through their impact on investment flows between currencies. Higher interest rates offer higher returns on investments in that currency, thus attracting foreign capital that appreciates the currency's value. Inflation rates are also critical as they affect purchasing power and can lead to monetary policy changes, directly impacting forex markets. A stable, low inflation rate is typically a sign of a healthy economy and positively affects the currency's value [11]. Unemployment rates provide insights into the economic health of a country [12]. High unemployment rates can indicate economic distress, leading to weaker currency values, while low rates suggest robust economic activity and are bullish for the currency. The performance of major equity indices like the ASX200 for Australia and NZX50 for New Zealand also serves as a bellwether for economic sentiment and financial health. Movements in these indices can reflect investor confidence and impact currency strength due to the interconnectedness of stock and forex markets. Economic growth, measured by GDP, outlines the broadest picture of economic health and potential market movements. Together with specific indicators such as the balance of trade figures and economic output data, these fundamental indicators provide a comprehensive view of the forces that drive currency values and are indispensable in forex prediction models.

III. Methodology

A. Data Collection

The dataset used in this research spans from January 2004 to April 2023 for the EUR/USD, EUR/NZD, and AUD/NZD currency pairs, containing 5016 rows of data and 29 independent factors, including both fundamental and technical indicators. For the NZD/USD currency pair, the dataset spans from January 2010 to April 2023 and consists of 3458 rows, with the same 29 independent factors. This extensive timeframe was chosen to ensure that the dataset covers diverse market conditions, capturing both stable and volatile periods, which enhances the robustness of predictive models.

This dataset is proprietary and accessible via a paid subscription to Thomson Reuters, ensuring high reliability and accuracy. The OHLC (Open, High, Low, and Close) data is provided in one day, making it suitable for capturing daily Forex price movements. The comprehensive selection of technical and fundamental indicators provides a complete market view, making the dataset ideal for time-series models that benefit from multi-faceted inputs.

Fundamental economic data were sourced from authoritative institutions to ensure data quality. The United States economic indicators were obtained from the Federal Reserve and the Federal Reserve Economic Data (FRED) database, serving as critical variables for analysing USD currency pairs [13], [14]. The European Union's economic indicators were accessed from Eurostat and the European Central Bank, offering insights into the Eurozone's financial conditions [15], [16]. For the New Zealand dollar, relevant economic data came from the Reserve Bank of New Zealand and Statistics New Zealand, providing a nuanced view of the NZD's influencing factors [17], [18]. Australian economic data, acquired from the Reserve Bank of Australia, similarly provide context for AUD movements [19].

Index closing prices for significant indices such as NZX50, EU50, ASX200, and S&P500, as well as commodities like gold and oil and the DXY index, were retrieved from TradingView to contextualise currency performance within broader market trends [20]. These indices reflect economic sentiment and help capture the interconnected influences of global markets on Forex prices.

Using the TA-Lib library, known for its strong technical analysis capabilities, additional technical indicators were generated from the OHLC data [21]. This process included the calculation of ATR, ROC_7, RSI_14, EMA_50, and Bollinger Bands, applying industry-standard methods to enrich the dataset for advanced technical analysis.

The chosen dataset is integral to this research as it includes a diverse and comprehensive set of inputs for ML and DL models to analyse and predict Forex price movements. Its breadth of technical and fundamental indicators supports models like LSTM and XGBoost in capturing both short-term price dynamics and long-term economic influences, making it ideally suited for developing accurate Forex prediction models. The dataset was partitioned as follows: 75% for training to capture market dynamics, 15% for validation to fine-tune parameters and prevent overfitting, and 10% for testing to evaluate predictive performance, ensuring an unbiased and robust assessment of model accuracy.

B. Model Development

The evolution from raw data to actionable insights was facilitated by applying sophisticated machine learning models, each selected for its specialised capability to decode the complexities of forex market data.

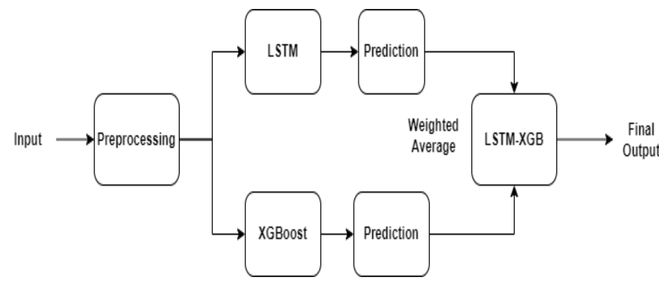

Fig. 1. Ensemble model combining LSTM and XGBoost, where LSTM captures temporal patterns, and XGBoost handles structured data for improved Forex prediction.

1) LSTM: LSTM networks, ideal for time series data, were applied to technical indicators (LSTM_TM), fundamental indicators (LSTM_FM), and a combination of both (LSTM_TM_FM). To improve generalisation, the LSTM models comprised two LSTM layers with `Dropout` and `BatchNormalization` layers. Each LSTM model was configured with hyperparameters, including a tuned number of layers, units per layer, learning rate, and batch size. Key optimisations included L2 regularization, `ReduceLROnPlateau` for learning rate adjustments, and early stopping to mitigate overfitting.

2) XGBoost: XGBoost was chosen for its efficiency in handling structured data, deployed on technical (XGB_TM), fundamental (XGB_FM), and combined indicators (XGB_TM_FM). Hyperparameter tuning was conducted using `GridSearchCV` with 5-fold cross-validation, optimising parameters such as `max_depth`, `learning_rate`, `n_estimators`, `subsample`, and `colsample_bytree`. The objective function was set to `reg:squarederror` to minimize prediction error.

3) Ensemble Model: The Ensemble model integrates LSTM and XGBoost predictions using a weighted averaging technique visualised in Figure 1. Each model was first trained separately, and weights were assigned based on performance metrics, prioritising the LSTM for its temporal analysis capabilities. This configuration leverages LSTM's ability to capture long-term dependencies and XGBoost's strength in identifying complex patterns in structured data. Performance metrics, including MAE, MAPE, RMSE, and R^2, were used to evaluate and fine-tune the ensemble.

4) Transformer Models: Two Transformer models were implemented:

- Transformer Encoder-Decoder (TFM_EN_DE_TM/FM/TM_FM): Configured with multi-head attention and tuned layers in both encoder and decoder, this model captures dependencies across technical and fundamental data sequences. Hyperparameters such as `num_layers`, `attention heads`, and learning rate were optimised to enhance sequential data interpretation.

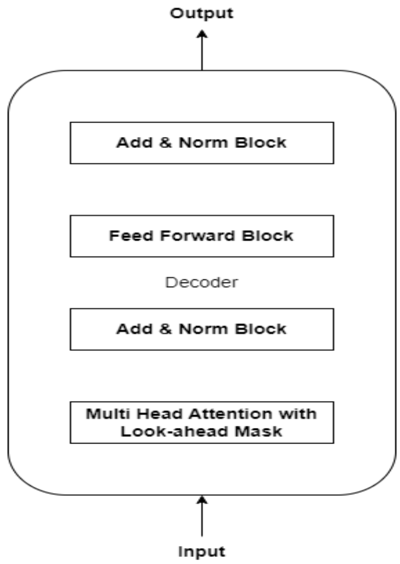

Fig. 2. Decoder architecture of the Transformer model, utilising attention mechanisms to capture long-range dependencies for forex prediction.

- Transformer Decoder-Only (TFM_DE_TM/FM/TM_FM):By leveraging decoder attention and look-ahead masking, this model excels in autoregressive tasks, as shown in figure 2, enabling it to predict future price movements solely based on historical data. A similar configuration of attention mechanisms and regularisation techniques was applied to ensure predictive robustness.

For all models, training was conducted with early stopping and learning rate scheduling to prevent overfitting and ensure model stability. The performance of each model was assessed using a suite of metrics to provide quantitative evaluation across multiple currency pairs.

IV. Results And Analysis

Building on our methodology, we've prioritized the RMSE in our Results section as the definitive measure for model comparison. RMSE is a rigorous standard because it quantifies prediction errors, directly reflecting a model's accuracy. A lower RMSE score indicates closer predicted values to the actual market prices, which is essential for reliable forex market analysis. This focus aligns with our objective to highlight models that not only perform well on historical data but also promise precision in real-world forecasting scenarios.

A. Fundamental Indicators

Focusing on models that utilized fundamental indicators in isolation revealed insightful outcomes. The XGBoost and Ensemble approaches, in particular, stood out for their notable accuracy, suggesting that fundamental data alone can be significantly predictive of forex price movements. The comparative analysis of models underscores the intrinsic value of economic indicators in forecasting currency values.

TABLE II
Models Using Fundamental Indicators with OHLC One-Day Time Frame

Model	Currency Pair	MAE	MAPE	RMSE	R^2
LSTM_FM	EUR/USD	0.1097	1.2457%	0.0190	0.9079
	NZD/USD	0.0705	0.7480%	0.0059	0.9630
	EUR/NZD	0.0647	0.8150%	0.0165	0.8485
	AUD/NZD	0.0715	0.5889%	0.0075	0.9165
XGB_FM	EUR/USD	0.0077	0.7379%	0.0157	0.9363
	NZD/USD	0.0068	1.0679%	0.0093	0.9525
	EUR/NZD	**0.0051**	**0.2982%**	**0.0068**	**0.9844**
	AUD/NZD	**0.0023**	**0.2124%**	**0.0029**	**0.9845**
ENS_LSTM-XGB_FM	**EUR/USD**	**0.0101**	**0.9396%**	**0.0155**	**0.9384**
	NZD/USD	**0.0046**	**0.7017%**	**0.0057**	**0.9825**
	EUR/NZD	0.0061	0.3566%	0.0078	0.9796
	AUD/NZD	0.0029	0.2688%	0.0036	0.9751
TFM_EN_DE_FM	EUR/USD	0.0290	2.7551%	0.0379	0.6250
	NZD/USD	0.0176	2.8499%	0.0226	0.4642
	EUR/NZD	0.0268	1.5957%	0.0345	0.3354
	AUD/NZD	0.0169	1.5886%	0.0204	0.3792
TFM_DE_FM	EUR/USD	0.0253	2.3743%	0.0292	0.7794
	NZD/USD	0.0139	2.2515%	0.0178	0.6694
	EUR/NZD	0.0228	1.3748%	0.0293	0.5187
	AUD/NZD	0.0135	1.2685%	0.0171	0.5614

Table II reveals that the Ensemble LSTM-XGB_FM model achieved the lowest RMSE for the EUR/USD and NZD/USD pairs, whereas the XGB_FM model was most accurate for the EUR/NZD and AUD/NZD pairs.

TABLE III
Models Using Technical Indicators with OHLC One-Day Time Frame

Model	Currency Pair	MAE	MAPE	RMSE	R^2
LSTM_TM	EUR/USD	0.0415	0.4635%	0.0065	0.9888
	NZD/USD	0.0592	0.6291%	0.0050	0.9738
	EUR/NZD	0.0341	0.4273%	0.0092	0.9531
	AUD/NZD	0.0318	0.2619%	0.0037	0.9796
XGB_TM	EUR/USD	0.0077	0.7379%	0.0157	0.9363
	NZD/USD	0.0025	0.4079%	0.0048	0.9872
	EUR/NZD	**0.0032**	**0.1846%**	**0.0048**	**0.9920**
	AUD/NZD	**0.0013**	**0.1258%**	**0.0017**	**0.9943**
ENS_LSTM-XGB_TM	**EUR/USD**	**0.0045**	**0.4184%**	**0.0064**	**0.9893**
	NZD/USD	**0.0030**	**0.4605%**	**0.0039**	**0.9917**
	EUR/NZD	0.0037	0.2154%	0.0051	0.9913
	AUD/NZD	0.0016	0.1519%	0.0021	0.9917
TFM_EN_DE_TM	EUR/USD	0.0222	2.0422%	0.0255	0.8295
	NZD/USD	0.0121	2.0033%	0.0181	0.6555
	EUR/NZD	0.0114	0.6832%	0.0148	0.8771
	AUD/NZD	0.0039	0.3565%	0.0049	0.9644
TFM_DE_TM	EUR/USD	0.0172	1.6682%	0.0259	0.8246
	NZD/USD	0.0126	2.0856%	0.0176	0.6757
	EUR/NZD	0.0103	0.6152%	0.0132	0.9032
	AUD/NZD	0.0041	0.3741%	0.0051	0.9615

B. Technical Indicators

The performance evaluation of models relying solely on technical indicators underscored the robust predictive capabilities of the XGBoost and Ensemble models. These models demonstrated exceptional precision and reliability in their forecasts, as evidenced by their low RMSE values and high R^2 scores, underscoring their effectiveness in capturing and predicting market trends.

Table III reveals that ENS_LSTM-XGB_TM model achieved the lowest RMSE for both the EUR/USD and NZD/USD pairs. Similarly, the XGB_TM model demonstrated superior performance for the EUR/NZD and AUD/NZD pairs with the lowest RMSE scores.

C. Technical and Fundamental Indicators

The integration of technical and fundamental indicators yielded mixed results. Specific models, notably XGB_TM_FM, showed a remarkable ability to maintain high accuracy levels. This indicates that adding fundamental data introduces complexity but also potentially enhances model performance when effectively harnessed.

Table IV reveals that the Ensemble LSTM-XGB_TM_FM model achieved the lowest RMSE for the EUR/USD and NZD/USD pairs, highlighting its effectiveness in leveraging technical and fundamental indicators for accurate predictions. Similarly, the XGB_TM_FM model demonstrated the lowest RMSE for the EUR/NZD and AUD/NZD pairs, indicating robust performance across these currency pairs when integrating technical and fundamental data.

TABLE IV
MODELS USING TECHNICAL AND FUNDAMENTAL INDICATORS WITH OHLC ONE-DAY TIME FRAME

Model	Currency Pair	MAE	MAPE	RMSE	R^2
LSTM_TM_FM	EUR/USD	0.1140	1.2640%	0.0159	0.9341
	NZD/USD	0.0931	1.0096%	0.0081	0.9306
	EUR/NZD	0.0476	0.5979%	0.0123	0.9153
	AUD/NZD	0.0788	0.6518%	0.0082	0.9004
XGBoost_TM_FM	EUR/USD	0.0077	0.7379%	0.0157	0.9363
	NZD/USD	0.0042	0.6703%	0.0062	0.9794
	EUR/NZD	**0.0046**	**0.2708%**	**0.0061**	**0.9868**
	AUD/NZD	**0.0020**	**0.1888%**	**0.0025**	**0.9879**
ENS_LSTM-XGB_TM_FM	**EUR/USD**	**0.0092**	**0.8477%**	**0.0118**	**0.9641**
	NZD/USD	**0.0037**	**0.5701%**	**0.0052**	**0.9856**
	EUR/NZD	0.0049	0.2899%	0.0064	0.9859
	AUD/NZD	0.0026	0.2420%	0.0032	0.9805
TFM_EN_DE_TM_FM	EUR/USD	0.0229	2.1760%	0.0306	0.7547
	NZD/USD	0.0127	2.0954%	0.0184	0.6440
	EUR/NZD	0.0280	1.6832%	0.0333	0.3814
	AUD/NZD	0.0126	1.1768%	0.0161	0.6108
TFM_DE_TM_FM	EUR/USD	0.0201	1.8884%	0.0264	0.8200
	NZD/USD	0.0145	2.3681%	0.0179	0.6677
	EUR/NZD	0.0208	1.2425%	0.0250	0.6494
	AUD/NZD	0.0095	0.8929%	0.0130	0.7493

D. Evaluation of Non-Stationarity and Benchmark Comparison

The Augmented Dickey-Fuller (ADF) test, a crucial analytical tool for examining stationarity in time-series data, was applied to the EUR/USD dataset, serving as a representative sample for the forex data used in this study. The test examines whether the time series contains a unit root, which indicates non-stationarity.

Null Hypothesis (H_0): The time series has a unit root, indicating non-stationarity.

$$H_0 : \delta = 0 \text{ in } \Delta Y_t = \alpha + \beta t + \delta Y_{t-1} + \epsilon_t$$

where:

- ΔY_t: first difference of the time series at time t,
- α: constant term,
- βt: deterministic trend,
- δ: coefficient on the lagged level of the series,
- Y_{t-1}: lagged level of the series,
- ϵ_t: error term at time t.

The ADF statistic was -1.831319, which did not meet the critical threshold values of -3.432, -2.862, and -2.567 for the 1%, 5%, and 10% confidence levels, respectively. Additionally, a high p-value of 0.365027 strongly indicates non-rejection of the null hypothesis, affirming the presence of a unit root and thereby confirming the non-stationarity of the data.

Given these findings, the Random Walk with Drift model was chosen as a baseline for comparison, as it is well-suited to handle non-stationary data. The Random Walk with Drift model assumes that future prices depend on past prices plus a constant drift and a random error term, represented as:

$$P_t = P_{t-1} + \mu + \epsilon_t$$

where:

- P_t: price at time t,
- P_{t-1}: price at time $t-1$,

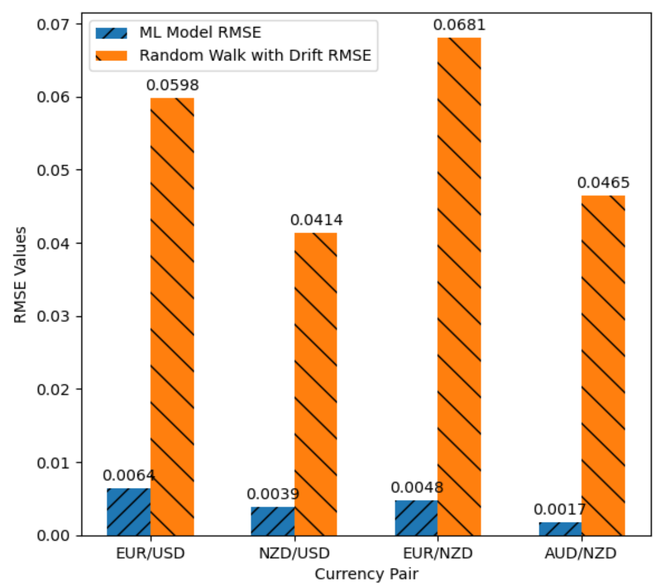

Fig. 3. Comparison of RMSE values for LSTM, XGBoost, Transformer, and Random Walk with Drift across currency pairs, showing the model's superior performance.

Figure 3 illustrates the RMSE values for various machine learning and deep learning (ML/DL) models compared to the Random Walk with Drift model across several major currency pairs. The results show that the ML/DL models consistently outperform the Random Walk model, particularly in capturing the intricate patterns of forex market dynamics affected by non-stationarity.

The significantly lower RMSE values for our ML/DL models compared to the Random Walk with Drift demonstrate the enhanced predictive accuracy of these approaches. These results underscore the importance of using advanced models, such as LSTM, XGBoost, and Transformers, which are better equipped to handle non-stationary data in forex prediction compared to simpler baselines like the Random Walk model.

V. Conclusion

We have made significant strides in bridging finance, machine learning, and deep learning, particularly in forex price prediction. The findings, as evidenced in Tables II, III, and IV,indicate that the decoder-only architecture of Transformers exhibits promising results in forex prediction, outperforming the encoder-decoder framework. Moreover, our comparative analysis, as depicted in Figure 4,has demonstrated that technical models outperform those based on fundamental indicators or a combination of both. This could be attributed to the direct reflection of market sentiments and patterns captured in technical data, which these models distill and interpret.

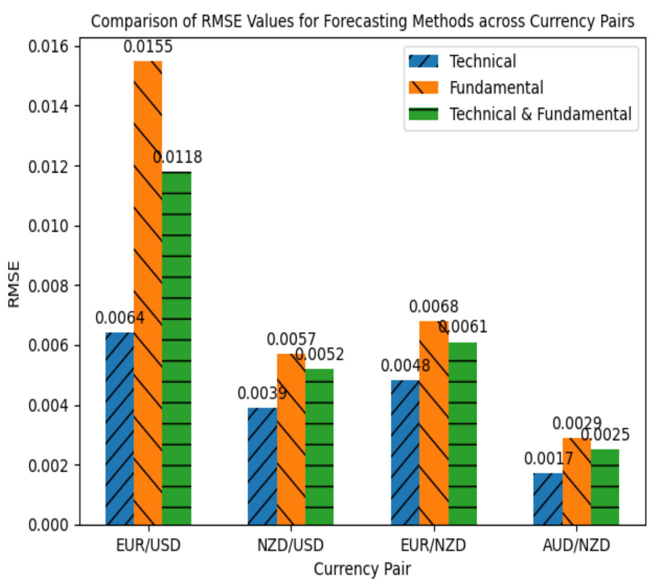

Fig. 4. Comparison of RMSE values across currency pairs using LSTM, XGBoost, and ensemble models with technical, fundamental, and combined indicators.

The intricacies of forex prediction are visually encapsulated in Figure 4, which provides a comparative analysis of RMSE values across various forecasting models for key currency pairs. This figure serves as a cornerstone in our discussion, underlining the strengths and limitations of different predictive approaches in the context of forex market dynamics.

Most compelling is the revelation from Figure 1, employing ensemble methods, integrating the strengths of XGBoost and LSTM models through a weighted average technique, significantly enhancing model robustness and generalizability. Such ensemble models have shown superior performance in our study, underpinning their potential as a robust framework for future forex market analysis.

References

[1] N. Rundo, N. Trenta, N. Di Stallo, and N. Battiato, “Machine Learning for Quantitative Finance Applications: A survey,” Applied Sciences, vol. 9, no. 24, p. 5574, Dec. 2019, doi: 10.3390/app9245574.

[2] L. Ni, Y. Li, X. Wang, J. Zhang, J. Yu, and C. Qi, “Forecasting of Forex Time Series Data Based on Deep Learning,” Procedia Computer Science, vol. 147, pp. 647–652, 2019, doi: https://doi.org/10.1016/j.procs.2019.01.189.

[3] A. J. Dautel, W. K. Härdle, S. Lessmann, and H.-V. Seow, “Forex exchange rate forecasting using deep recurrent neural networks,” Digital Finance, vol. 2, no. 1–2, pp. 69–96, Mar. 2020, doi: 10.1007/s42521-020-00019-x.

[4] L. Ni, Y. Li, X. Wang, J. Zhang, J. Yu, and C. Qi, “Forecasting of Forex Time Series data based on deep learning,” Procedia Computer Science, vol. 147, pp. 647–652, Jan. 2019, doi: 10.1016/j.procs.2019.01.189.

[5] D. C. Yıldırım, I. H. Toroslu, and U. Fiore, “Forecasting directional movement of Forex data using LSTM with technical and macroeconomic indicators,” Financial Innovation, vol. 7, no. 1, Jan. 2021, doi: 10.1186/s40854-020-00220-2.

[6] J. Wang, X. Wang, J. Li, and H. Wang, “A prediction model of CNN-TLSTM for USD/CNY exchange rate prediction,” IEEE Access, vol. 9, pp. 73346–73354, Jan. 2021, doi: 10.1109/access.2021.3080459.

[7] Z. Hu, Y. Zhao, and M. Khushi, “A Survey of Forex and Stock Price Prediction Using Deep Learning,” Applied System Innovation, vol. 4, no. 1, p. 9, Feb. 2021, doi: https://doi.org/10.3390/asi4010009.

[8] M. Ayitey Junior, P. Appiahene, O. Appiah, and C. N. Bombie, “Forex market forecasting using machine learning: Systematic Literature Review and meta-analysis,” Journal of Big Data, vol. 10, no. 1, Jan. 2023, doi: https://doi.org/10.1186/s40537-022-00676-2.

[9] Md. S. Islam and E. Hossain, “Foreign exchange currency rate prediction using a GRU-LSTM hybrid network,” Soft Computing Letters, vol. 3, p. 100009, Dec. 2021, doi: 10.1016/j.socl.2020.100009.

[10] P. H. Vuong, T. T. Dat, T. K. Mai, P. H. Uyen, and P. Bao, “Stock-Price forecasting based on XGBOOST and LSTM,” Computer Systems Science and Engineering, vol. 40, no. 1, pp. 237–246, Jan. 2022, doi: 10.32604/csse.2022.017685.

[11] G. Adler, K. S. Chang, and Z. Wang, “Patterns of foreign exchange intervention under inflation targeting,” Latin American Journal of Central Banking, vol. 2, no. 4, p. 100045, Dec. 2021, doi: 10.1016/j.latcb.2021.100045.

[12] D. Bathia, R. Demirer, R. Gupta, and K. Kotzé, “Unemployment fluctuations and currency returns in the United Kingdom: Evidence from over one and a half century of data,” Journal of Multinational Financial Management, vol. 61, p. 100679, Sep. 2021, doi: 10.1016/j.mulfin.2021.100679.

[13] Federal Reserve Economic Data, “Economic Data,” available at: https://fred.stlouisfed.org/, accessed on: July 26, 2023.

[14] Federal Reserve, “Federal Reserve,” [Online]. Available: https://www.federalreserve.gov/. Accessed on: July 19, 2023.

[15] European Central Bank, “Statistics,” available at: https://www.ecb.europa.eu/stats/policy_and_exchange_rates, accessed on: July 19, 2023.

[16] Eurostat, “Data and Statistics,” available at: https://ec.europa.eu/eurostat, accessed on: July 26, 2023.

[17] Reserve Bank of New Zealand, “Statistics,” available at: https://www.rbnz.govt.nz/statistics, accessed on: July 24, 2023.

[18] Statistics New Zealand, “Data and Statistics,” available at: https://www.stats.govt.nz/, accessed on: July 25, 2023.

[19] Reserve Bank of Australia, “Statistics,” available at: https://www.rba.gov.au/statistics/, accessed on: March 12, 2024.

[20] TradingView, “Comprehensive Financial Charts for Analysis,” [Online]. Available: https://www.tradingview.com/. Accessed on: January 16, 2024.

[21] TA-Lib, “Technical Analysis Library,” [Online]. Available: https://ta-lib.org/. Accessed on: September 28, 2023.